Question

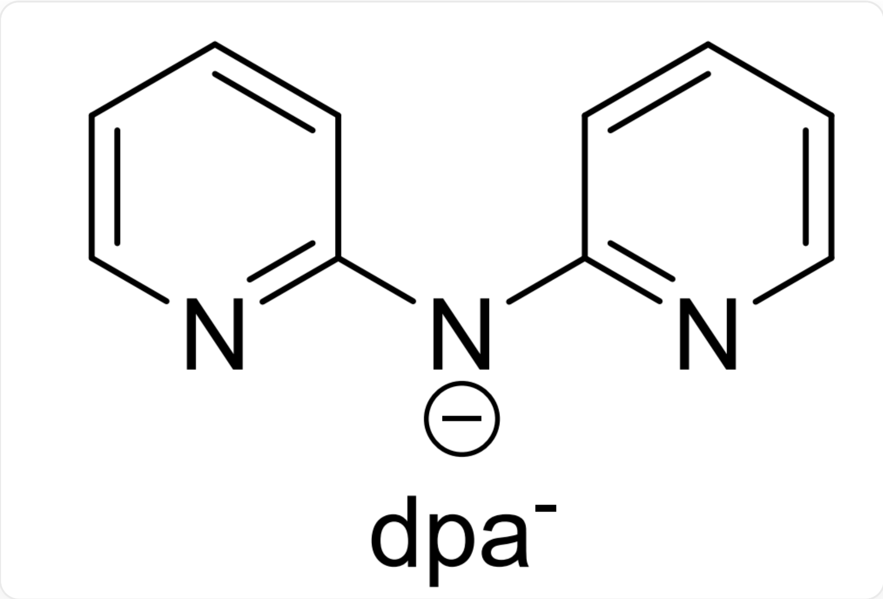
To a mixed CH_2Cl_2 solution of $[Au(PPh_3)_2][SbF_6]$ and $Au \cdot \mathbf{dpa}$ (the structure of the ligand $\mathbf{dpa}^-$ is shown below), a methanol solution of CH_3ONa was added. The mixture was stirred at room temperature, and the system turned into a green solution. Under vigorous stirring, a freshly prepared ethanol solution of $NaBH_4$ was gradually added dropwise to the system, and the solution gradually turned light brown, and finally turned dark brown. The reaction was continued with stirring in the dark for $20h$. The system was evaporated to dryness, washed with diethyl ether and ethanol to obtain a black solid $\mathbf{A}$. Single crystals of $\mathbf{A}$ were obtained by recrystallization of $\mathbf{A}$ and studied. The following partial information about $\mathbf{A}$ was obtained:

(a) $\mathbf{A}$ is a 1:2 type salt, and its cation is a cluster ion centered on the gold atom cluster Au_x^{y+}; Au_x^{y+} has a double-layer nested structure in the form of $[Au_a@Au_b]$.

(b) The main signal of the cation of $\mathbf{A}$ in time-of-flight mass spectrometry is $m/z = 4711.02$

(c) $\mathbf{A}$ thermally decomposes at $950^{\circ}C$ with a weight loss of 36.5% , and the remaining solid residue is Au element.

(d) Partial elemental content (mass fraction) of $\mathbf{A}$: $C, 24.8\%; N, 2.55\%; P, 2.50\%$



The figure shows the structure of $\mathbf{dpa}^-$, and the smiles is expressed as: C1([N-]C2=NC=CC=C2)=NC=CC=C1

There are the following statements:

1, $|\frac{x-y}{x+2y}|^{\ln|\frac{x+y}{x-y}|} = 0.70$.

2, The molecular weight of $\mathbf{A}$ is 9657.8.

3, $|\frac{a-b}{a+2b}|^{\ln|\frac{a+b}{a-b}|} = 0.07$.

4, The phosphorus atoms in the cation of $\mathbf{A}$ are coordinated to the inner gold atoms.

Then the following options contain all the correct options (the error of 1, 3 within 0.01 is considered correct):

A. All other options are incorrect

B. 1

C. 2

D. 3

E. 4

F. 1, 2

G. 1, 3

H. 1, 4

I. 2, 3

J. 2, 4

K. 3, 4

L. 1, 2, 3

M. 1, 2, 4

N. 1, 3, 4

O. 2, 3, 4

P. 1, 2, 3, 4

Answer

Correct Answer: G

Detailed Explanation

First, calculate the ratio of atoms by mass fraction:

$$\mathrm{C} : \mathrm{N} : \mathrm{P} = \frac{24.8\%}{12.01} : \frac{2.55\%}{14.01} : \frac{2.50\%}{30.97} \approx 11.35 : 1 : 0.4435$$

The nitrogen atoms in $\mathbf{A}$ only come from the ligand $\mathbf{dpa}^-$, so it is a multiple of 3. Thus, the above formula is transformed into:

$$11.35 : 1 : 0.4435 \approx 34.05 : 3 : 1.331 \approx 102 : 9 : 4$$

Therefore, $\mathrm{C} : \mathrm{N} : \mathrm{P} = 102 : 9 : 4$. The total number of carbon atoms in 4 $\mathrm{PPh_3}$ and 3 $\mathbf{dpa}^-$ is $4 \times 18 + 3 \times 10 = 102$, which matches the calculation.

CHECKPOINT 1 PTS

Atomic ratio of $\mathbf{A}$ $\mathrm{C} : \mathrm{N} : \mathrm{P} = 102 : 9 : 4$

CHECKPOINT 2 PTS

The carbon-containing ligands of $\mathbf{A}$ are only $\mathrm{PPh_3}$ and $\mathbf{dpa}^-$ with a ratio of 4:3

The total molecular weight of 4 $\mathrm{PPh_3}$ and 3 $\mathbf{dpa}^-$ is $4 \times 262.3 + 3 \times 170.192 = 1559.8$.

$\mathbf{A}$ is a 1:2 type salt. Here, the anion can only be SbF_6^-, so the cation carries two positive charges.

CHECKPOINT 1 PTS

The cation of $\mathbf{A}$ carries two positive charges

Therefore, the mass spectrometry signal m/z = M/2, from which the molecular weight of the cation can be obtained as $2 \times 4711.02 = 9422.04$.

If the cation contains 4 PPh_3 and 3 $\mathbf{dpa}^-$, the number of Au is $\frac{9422.04-1559.8}{197} = 39.90$, which means the gold content is too high, inconsistent with the thermogravimetric results.

If the cation contains 8 PPh_3 and 6 $\mathbf{dpa}^-$, the number of Au is $\frac{9422.04-1559.8\times 2}{197} = 32.00$, which is consistent with the question. Therefore, the cation is $[Au_{32}(PPh_3)_8(dpa)_6]^{2+}, x = 32, y = 8$.

CHECKPOINT 2 PTS

The cation of $\mathbf{A}$ is $[Au_{32}(PPh_3)_8(dpa)_6]^{2+}$

CHECKPOINT 0.5 PTS

$x = 32, y = 8$

Thus, the chemical formula of $\mathbf{A}$ is $[Au_{32}(PPh_3)_8(dpa)_6][SbF_6]_2$, and the molecular weight is 9893.56.

CHECKPOINT 1 PTS

The molecular weight of $\mathbf{A}$ is 9893.56

Au_{32}^{8+} has a double-layer nested structure in the form of $[Au_a@Au_b]$. It is speculated that it is a nesting of two polyhedra. Considering the simplest polyhedron, i.e., a regular polyhedron, then 32 atoms can be exactly divided into 20+12, which is exactly a regular dodecahedron plus a regular icosahedron.

CHECKPOINT 1 PTS

32 atoms form a regular dodecahedron and a regular icosahedron

The inside is a regular icosahedron with fewer atoms, and the outside is a regular dodecahedron with more atoms. Therefore, $a = 12, b = 20$.

CHECKPOINT 1 PTS

$a = 12, b = 20$

Since the space inside the cage is very small, and PPh_3 has a large volume, it can only coordinate on the outside. Therefore, phosphorus atoms coordinate with the outer gold atoms.

CHECKPOINT 1 PTS

PPh_3 has a large volume, and phosphorus atoms coordinate with the outer gold atoms

The following is an analysis of the options:

1, $|\frac{x-y}{x+2y}|^{\ln|\frac{x+y}{x-y}|} = 0.70$. Correct.

2, The molecular weight of $\mathbf{A}$ is 9893.56. Incorrect.

3, $|\frac{a-b}{a+2b}|^{\ln|\frac{a+b}{a-b}|} = 0.07$. Correct.

4, In the cation of $\mathbf{A}$, phosphorus atoms coordinate with the outer gold atoms. Incorrect.

Choose G